Parktown Boys' High School

District: 9

Geography Department

JUNE EXAM

PAPER 1

Grade 12

Examiner: Ms A Clayton	Time: 3 Hour
Moderator: Mr A Meintjes	Total: 150 Marks
Number of pages: 17	Date: 1 June 2026

INSTRUCTIONS AND INFORMATION

1. This question paper consists of TWO SECTIONS.

   SECTION A
   - QUESTION 1: CLIMATE AND WEATHER (60)
   - QUESTION 2: GEOMORPHOLOGY (60)

   SECTION B
   - QUESTION 3: GEOGRAPHICAL SKILLS AND TECHNIQUES (30)
2. Answer ALL THREE questions.
3. All diagrams are included in the QUESTION PAPER.
4. Leave a line between the subsections of questions answered.
5. Start EACH question at the top of a NEW page.
6. Number the answers correctly according to the numbering system used in this question paper.
7. Do NOT write in the margins of the ANSWER BOOK.
8. Draw fully-labelled diagrams when instructed to do so.
9. Answer in FULL SENTENCES, except when you have to state, name, identify or list.
10. Units of measurement MUST be indicated in your final answer, e.g. 1 020 hPa, 14 °C and 45 m.
11. You may use a non-programmable calculator.
12. You may use a magnifying glass.
13. Write neatly and legibly.

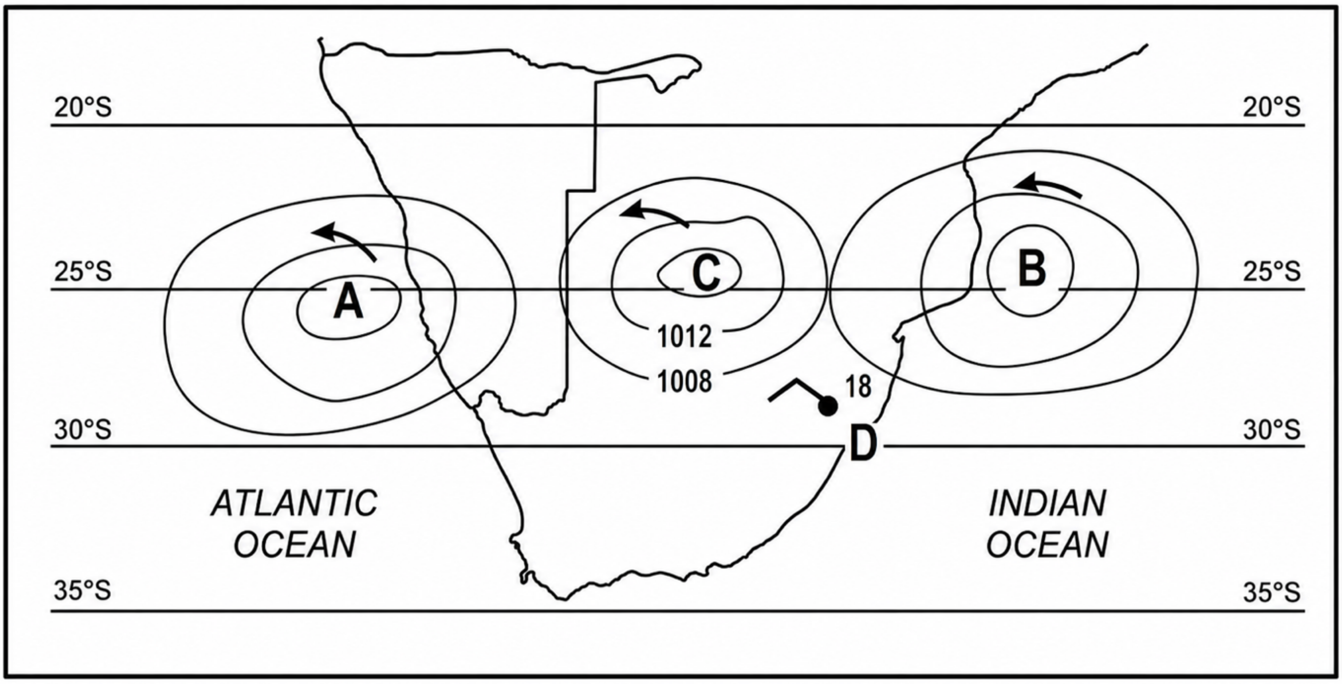
SPECIFIC INSTRUCTIONS AND INFORMATION FOR SECTION B

14. A 1: 50 000 topographical map of 2531BC MALELANE and a 1: 10 000 orthophoto map 2531 BC 8 MALELANE are provided.
15. The area demarcated in RED on the topographical map represents the area covered by the orthophoto map.
16. Marks will be allocated for steps in calculations.
17. You must hand in the topographical and the orthophoto maps to the invigilator at the end of this examination session.

SECTION A: CLIMATE AND WEATHER AND GEOMORPHOLOGY

QUESTION 1: CLIMATE AND WEATHER

1.1. Refer to the figure below which shows anticyclones over South Africa.



Choose the correct word(s) from those given in brackets. Write only the word(s) next to the question numbers (1.1.1 to 1.1.7) in your answer booklet.

1.1.1 Pressure cell A is situated further (north/south) in summer.

1.1.2 Pressure cell B is named the (South Atlantic/South Indian) High Pressure Cell.

1.1.3 When isobars are elongated away from pressure cell B they form a (ridge/trough) of high pressure.

1.1.4 The pressure reading at C is approximately (1 012 hPa/1 016 hPa).

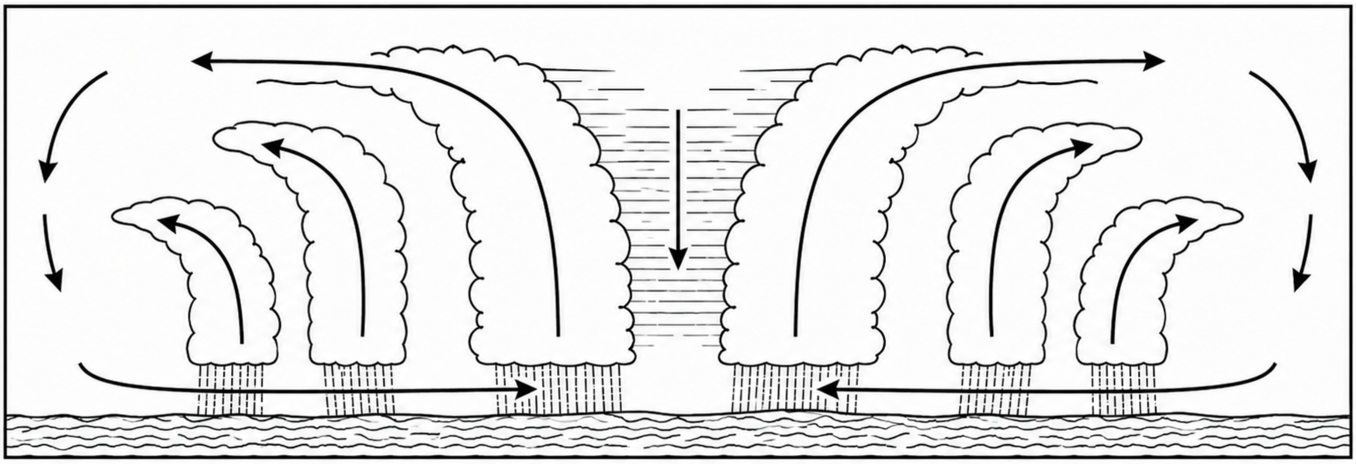
1.1.5 The wind speed at weather station D is (20 knots/10 knots).

1.1.6 The wind direction at weather station D is (north-east/north-west).

1.1.7 Pressure cells A, B and C represent the (equatorial low/subtropical high) pressure belt.

(7x1) (7)

1.2 Refer to the figure below which is a cross section of a Tropical Cyclone.



[Adapted from https://www.internetgeography.net/geotopics/what-is-the-structure-and-features-of-a-tropical-storm/]

Various options are provided as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.2.1–1.2.8) in the answer booklet, for example 1.2.9 A.

1.2.1 The cross section shows the … stage of a tropical cyclone.

A. formative
B. immature
C. mature
D. degenerating

1.2.2 The movement of the air around the low pressure is … in the northern hemisphere.

A. Clockwise
B. Anti-clockwise
C. West to east
D. North to south

1.2.3 The centre of the weather system is known as the … of the tropical cyclone.

A. cortex
B. hurricane
C. eye
D. moisture front

1.2.4 The reason for the calm weather in the centre of the tropical cyclone is …

A. subsiding air that warms adiabatically
B. subsiding air that cools adiabatically
C. ascending air that warms adiabatically
D. ascending air that cools adiabatically

1.2.5 The path of a tropical cyclone is from …

A. west to east
B. east to west
C. north to south
D. south to west

1.2.6 Which cloud type is most commonly associated with the stage of a tropical cyclone depicted in the cross section?

A. Cirrus
B. Stratus
C. Cumulonimbus
D. Altocumulus

1.2.7 Which process provides the main source of energy for a tropical cyclone?

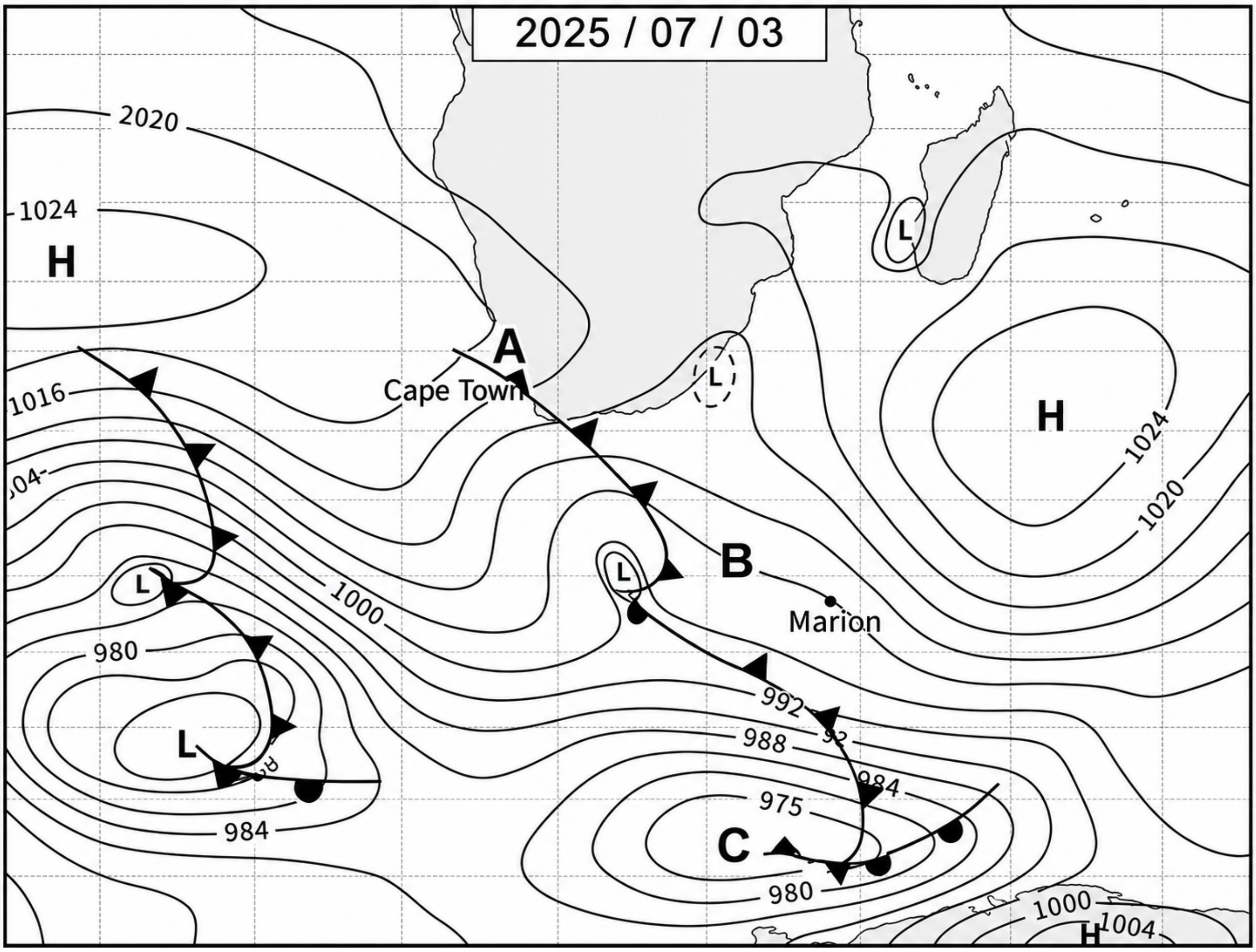
A. Conduction
B. Condensation of water vapour
C. Strong vertical wind shear
D. Friction from the ocean floor

1.2.8 In which part of a mature tropical cyclone will rainfall be the heaviest?

A. Above the ocean surface only
B. On the outer edge of the system
C. In the eye
D. In the eyewall

(8 x 1) (8)

1.3 Refer to the synoptic weather map below on mid-latitude cyclones.



1.3.1 What do we call a series of mid-latitude cyclones in succession?

(1 x 1) (1)

1.3.2 Name front A and sector B on the synoptic weather map.

(2 x 1) (2)

1.3.3 What influence does the South Indian Anticyclone, as shown on the synoptic weather map, have on the path followed by the mid-latitude cyclones?

(1 x 2) (2)

1.3.4 The mid-latitude cyclone at C is in the cold occlusion stage. Draw a labelled cross-section at C on the synoptic weather map. Marks will be awarded for the following:

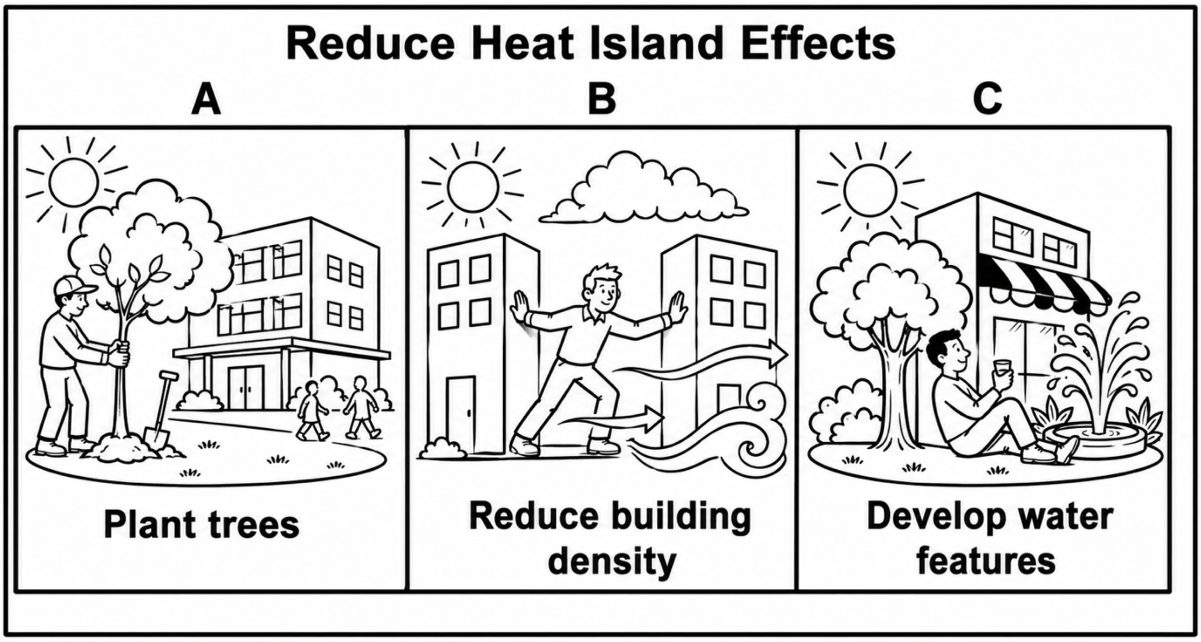
- Cross-section (1 x 1) (1)
- Cold air (1 x 1) (1)
- Cool air (1 x 1) (1)
- Warm sector (1 x 1) (1)

(4 x 1) (4)

1.3.5 Describe the changes in weather that Cape Town will experience as the mid-latitude cyclone passes over.

(3 x 2) (6)

1.4 Refer to the figure below which shows methods to reduce the effects of a heat island.



[Source: Examiner's own figure]

1.4.1 Define the term urban heat island.

(1 x 2) (2)

1.4.2 Which part of an urban area is likely to experience the highest temperatures? Choose the correct answer and write only the letter.

A. Rural-urban fringe
B. Central Business District
C. Nature reserve
D. Suburban area

(1 x 1) (1)

1.4.3 Describe how the glass windows in the urban area shown in the figure will result in the increase of temperature in the urban buildings.

(1 x 2) (2)

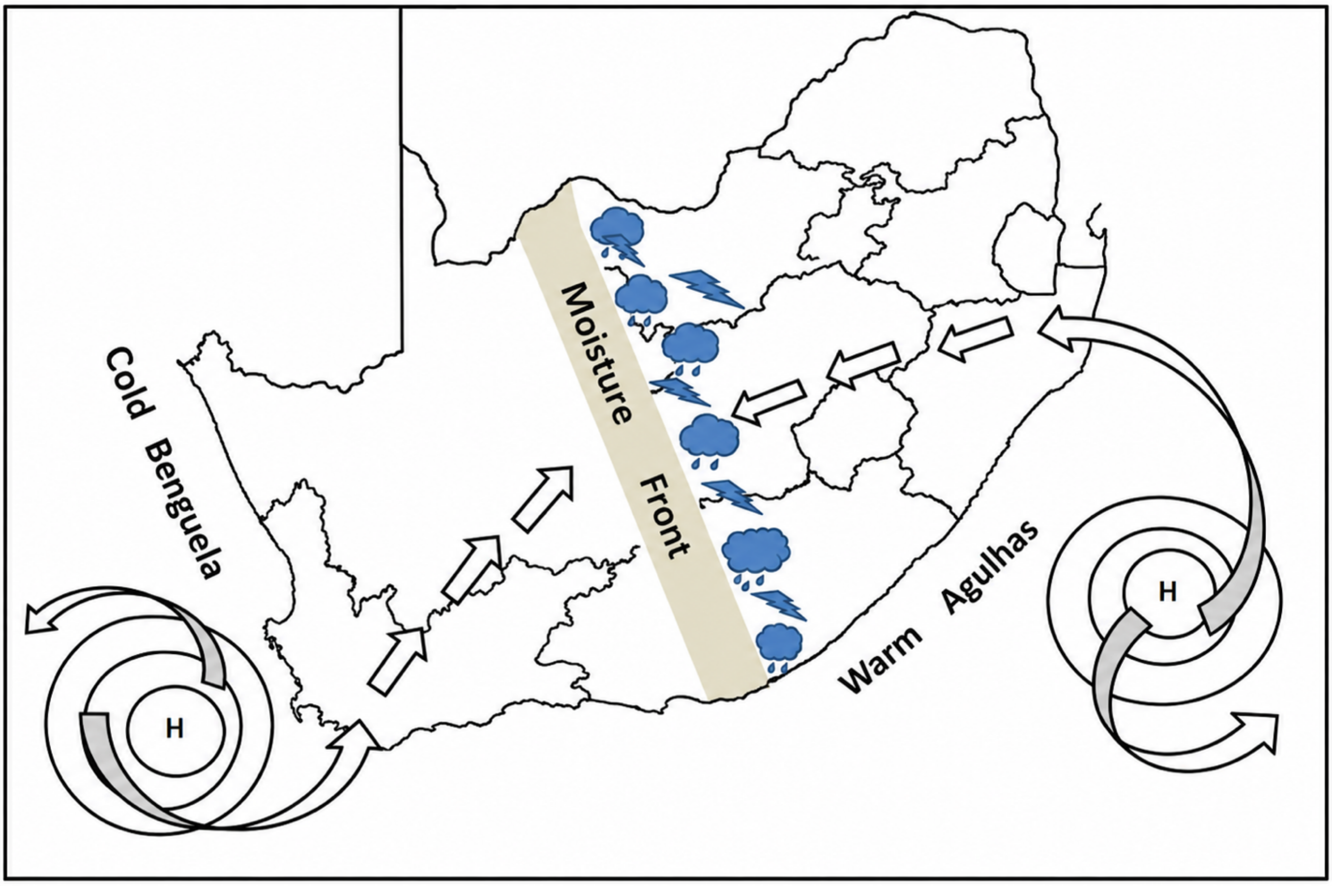
1.4.4 Explain TWO possible negative effects that urban heat islands have on the health of human beings.

(2 x 2) (4)

1.4.5 Explain how the THREE strategies (A, B and C) evident in the figure contribute to the reduction of the heat island effect.

(3 x 2) (6)

1.5 Refer to the figure below which shows a line thunderstorm.



[Adapted from: NSC Geography Prelim Paper 1 2021]

1.5.1 On which side of the moisture front do line thunderstorms occur? East or West? (1 x 1) (1)

1.5.2. Give a reason for your answer to QUESTION 1.5.1. (1 x 2) (2)

1.5.3 Describe the air that flows from the:

a. South Indian Anticyclone (1 x 2) (2)

b. South Atlantic Anticyclone (1 x 2) (2)

1.5.4 In a paragraph of approximately EIGHT lines, suggest the positive and negative impact that line thunderstorms have on the agricultural sector in the interior of South Africa. (4 x 2) (8)

[Question 1: 60 marks]

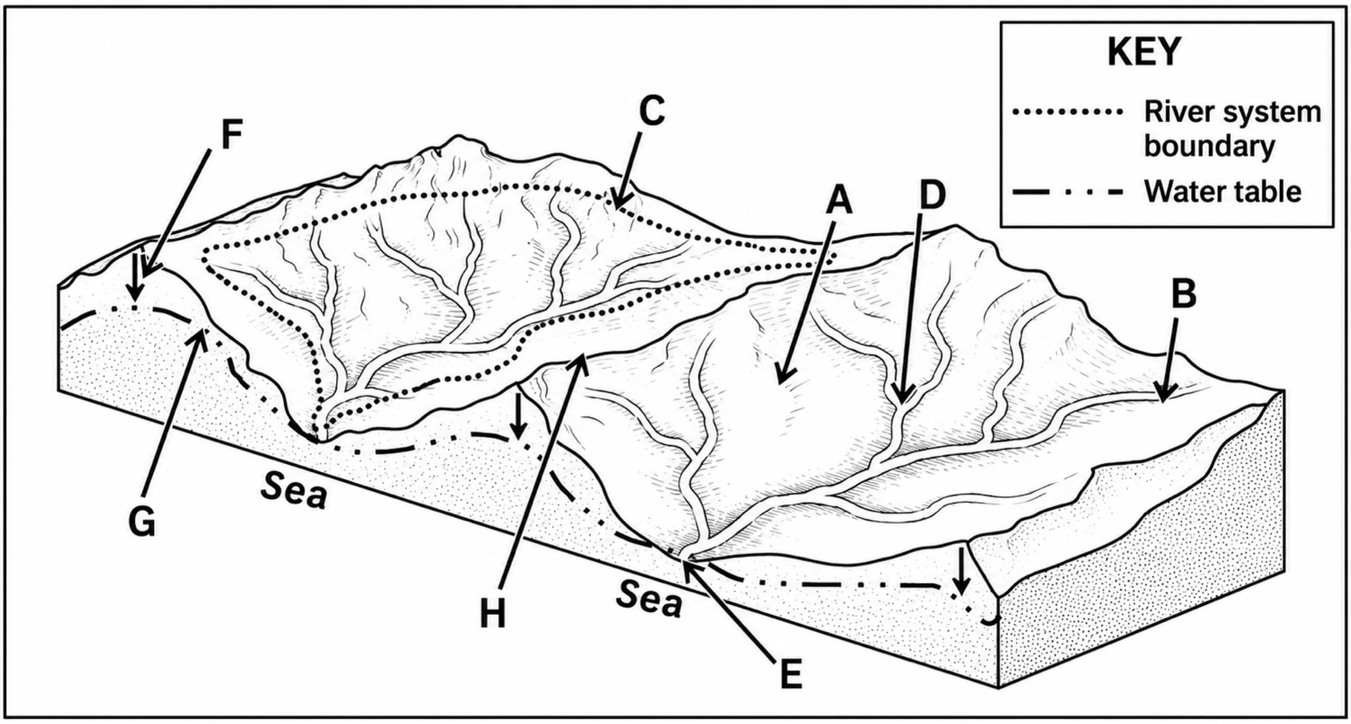
QUESTION 2: GEOMORPHOLOGY

2.1. Choose a term from COLUMN B that matches the fluvial landform description in COLUMN A. Write only the letter (A–H) next to the question numbers (2.1.1 to 2.1.7) in the answer booklet, e.g. 2.1.8 J.

Column A		Column B
2.1.1	Flat, natural feature next to a river	A rapid
2.1.2	An embankment along the river where coarse material is deposited first	B delta
		C meander
2.1.3	Curves or bends found along the course of a river	D braided stream
2.1.4	When a meander loop becomes separated from the river	E floodplain
2.1.5	Streams with multiple channels and islands of sediment between the channels	F oxbow lake
		G levee
2.1.6	A vertical drop in the course of a river as a result of softer rock eroding faster than hard rock	H waterfall
2.1.7	A depositional landform that occurs when a river flows into the ocean	

(7 x 1) (7)

2.2 Match the concepts below with the letters in the diagram. Write only the letter (A–H) next to the question numbers (2.2.1 to 2.2.8) in the answer booklet, e.g. 2.1.9 K.



[Source: Examiner's own diagram]

2.2.1 Source of the river

2.2.2 The water table

2.2.3 An interfluve

2.2.4 A drainage basin

2.2.5 The river mouth

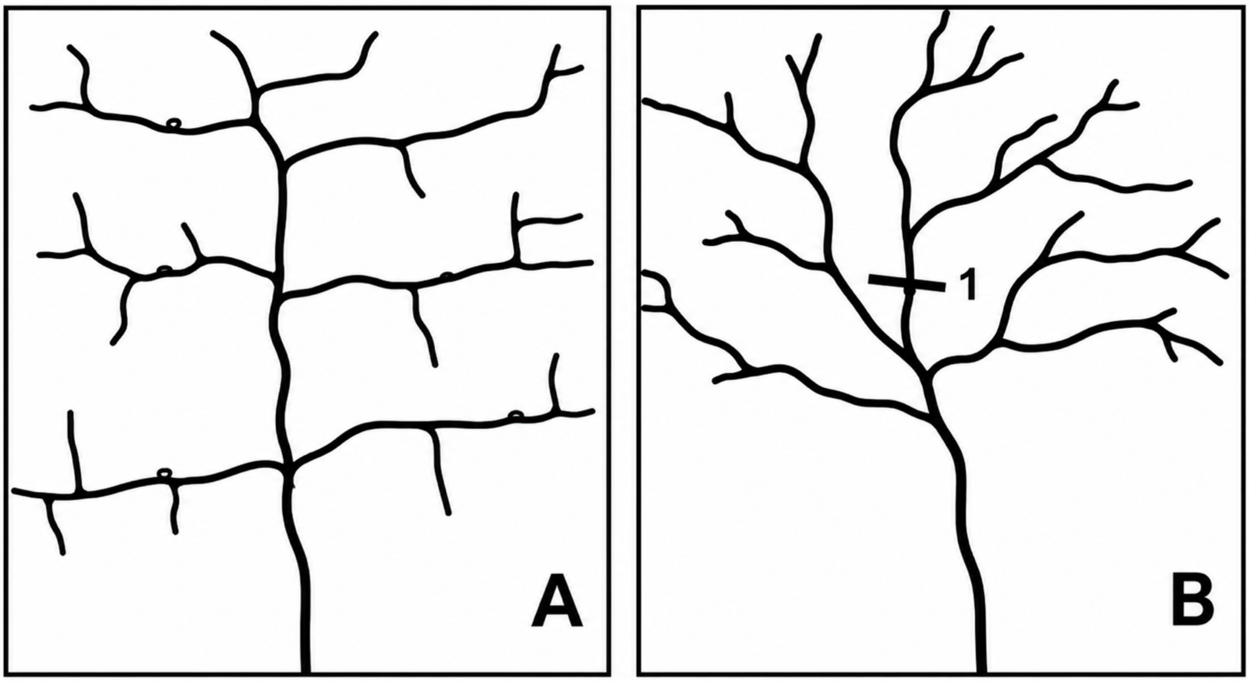
2.2.6 The watershed

2.2.7 A confluence

2.2.8 Process of infiltration

(8 x 1) (8)

2.3 Refer to the figure below depicting two different drainage patterns.



[Source: adapted from: https://www.planetary.org/space-images/drainage-patterns]

2.3.1 Identify drainage patterns A and B.

(2 x 1) (2)

2.3.2 Differentiate between the underlying rock structure of drainage patterns A and B respectively.

(2 x 2) (4)

2.3.3 Why are the tributaries of the main stream parallel to each other in drainage pattern A?

(1 x 2) (2)

2.3.4 Determine the stream order at point 1 in drainage pattern B.

(1 x 2) (2)

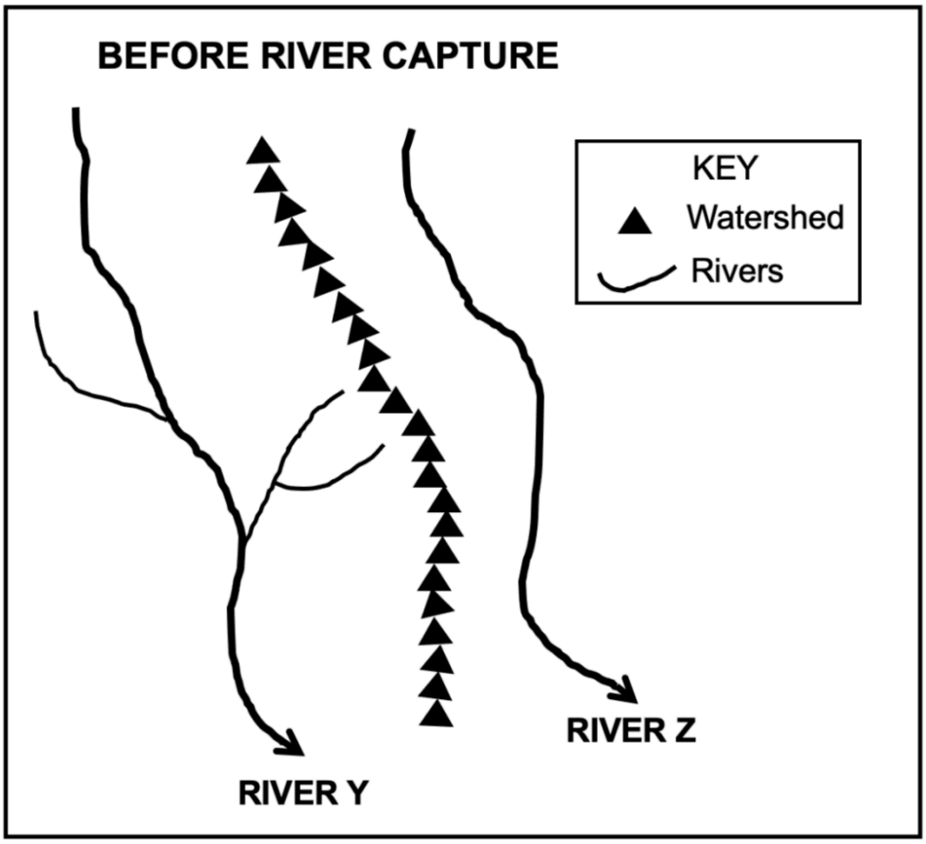
2.3.5 Choose the CORRECT word between brackets to make the statement TRUE.
The higher the stream order, the (higher/lower) the drainage density.

(1 x 1) (1)

2.3.6 Describe the relationship between:

a. Drainage density and low rainfall (1 x 2) (2)
b. Drainage density and steep gradient (1 x 2) (2)

2.4 Refer to the sketch map of rivers Y and Z before river capture has taken place.



[Adapted from NSC Geography P1 November 2021 Question Paper]

2.4.1 Define the concept river capture.

(1 x 2) (2)

2.4.2 State ONE condition needed for river capture to take place.

(1 x 2) (2)

2.4.3 Draw a sketch to illustrate the area after river capture has taken place.
Marks will be awarded for the accuracy of the sketch and indicating the following labels:

- Elbow of capture
- Misfit stream
- Wind gap

(1 + 3) (4)

2.4.4 Will river Y or Z experience rejuvenation after river capture?

(1 x 1) (1)

2.4.5 Give a reason for your answer to QUESTION 2.4.4.

(1 x 2) (2)

2.4.6 Refer to your answer to QUESTION 2.4.5 and explain the impact of the change on the captor stream.

(2 x 2) (4)

2.5 Study the extract on Gauteng River Catchment Management below.

Gauteng is located on the continental divide with some rivers flowing towards the Indian Ocean and others to the Atlantic Ocean. It contains the headwaters of a number of important river systems in an urban environment. The high flow velocity of the river causes erosion especially where vegetation cover is removed, or where the banks of rivers and streams are modified. Seasonal flooding is a real danger in several extensive areas in Gauteng. Canalisations of several rivers in urban as well as rural areas have further negative effects. Healthy riverbanks maintain the form of the river channel, provide habitat for species (aquatic and terrestrial) and filter sediment, minerals and light. Water quality includes the chemical, physical and bacteriological properties of water which determine its suitability for use. The urban nature of Gauteng (especially the central part), as well as road networks across the province, seals natural surfaces in a manner that does not allow natural infiltration of rainwater into the ground. This high runoff scenario during rainfall events coupled to pollution emanating from the urban environment puts a high level of stress on the river system of Gauteng.

[Source: https://armour.org.za/wp/about-us/armour-a-voice-for-water/]

2.5.1 Define the concept river management.

(1 x 2) (2)

2.5.2 Give ONE piece of evidence from the extract that shows the importance of catchment management in Gauteng.

(1 x 1) (1)

2.5.3 Discuss TWO economic impacts that that poor river management would have for Gauteng.

(2 x 2) (4)

2.5.4 In a paragraph of approximately EIGHT lines, provide sustainable strategies that can be implemented to deal with problems associated with poor river management systems.

(4 x 2) (8)

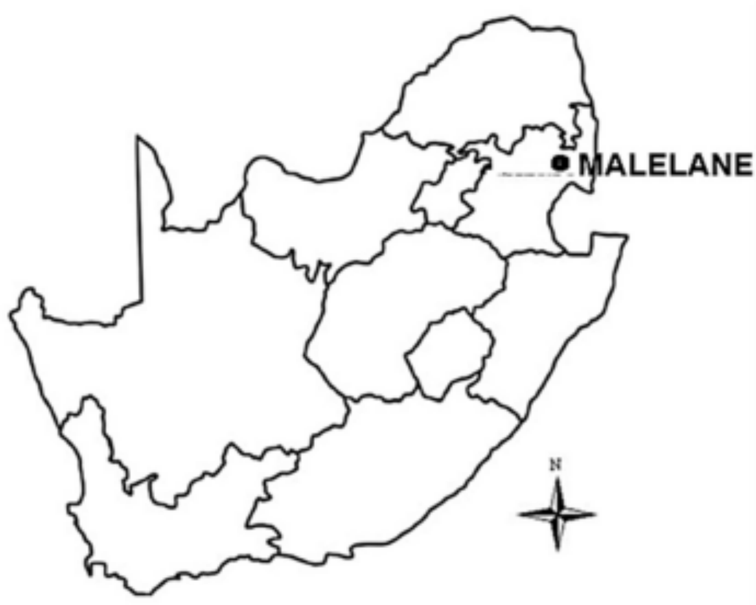
[Question 2: 60 marks]

Section A Total: 120 marks

SECTION B

QUESTION 3: GEOGRAPHICAL SKILLS AND TECHNIQUES

BACKGROUND INFROMATION ON MALELANE



Coordinates: 25°29'S ; 31°31'E

Malelane is a gateway to the Kruger National Park, Mozambique and Eswatini (formerly Swaziland). This charming farming town in Mpumalanga is situated on the N4 national highway about halfway between the capital city of Mpumalanga, Mbombela (formerly Nelspruit), and the capital city of Mozambique, Maputo.

Malelane sits on the intersection of three roads that lead to many of southern Africa's favourite tourist destinations. Driving into Malelane from Nelspruit, you can see the hilly outcrop directly ahead, with the Mozambique border post nestled at its base. Turn left and the promise of a few glorious days in the Kruger National Park beckons you. Turn right and the lush, tropical landscape of Eswatini lies on the horizon.

Malelane, features a hot, subtropical climate with around 547mm of annual rainfall. Temperatures are high, typical of the Lowveld, with, for example, January averaging around 26°C and winter days remaining warm.

[Source: https://hamiltonslodge.co.za/history-of-malelane/]

3.1 MAP SKILLS AND CALCULATIONS

Refer to the topographical map and the orthophoto map.

3.1.1 In which province is Malelane located?

A. Gauteng
B. Limpopo
C. Mpumalanga
D. KwaZulu-Natal

(1 x 1) (1)

3.1.2 The true bearing from spot height 557 in **E5** to spot height 522 in **E4** on the topographic map is:

A. 62°
B. 78°
C. 282°
D. 298°

(1 x 1) (1)

3.1.3 Using your answer in QUESTION 3.1.2, calculate magnetic bearing.

Magnetic Mearing = True Bearing + Magnetic Declination

(2 x 1) (2)

3.1.4 Refer to the large perennial water body (dam) in **B4** on the topographic map. Calculate the area covered by the dam in **B4** if the length is 1cm and the breadth is 0,8cm. Give the answer in km². Show ALL calculations.

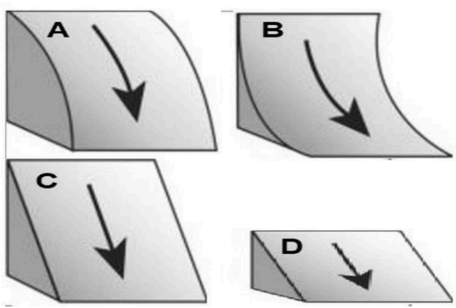
Area = length x breadth

(3 x 1) (3)

3.1.5 Choose the correct answer from the options provided below. Write only the letter (A – D) next to the question number.

A cross section of the slope between **8** in block **D4** and **9** in block **D3** on the orthophoto map extract is ....

(1 x 1) (1)



3.1.6 Give a reason for your answer to QUESTION 3.1.5.

(1 x 2) (2)

3.2 MAP INTERPRETATON

3.2.1 Much of the agricultural area surrounding Malelane is in the valley of the Umgwenya river.

(a) Choose the correct answer from the options provided below. Write only the letter (A – D) next to the question number.

The nocturnal (night-time) wind that develops during the night in this area is called a ... wind.

A. berg
B. föhn
C. anabatic
D. katabatic

(1 x 1) (1)

(b) Explain how the wind identified in QUESTION 3.2.1(a) promotes the formation of dense fog at **K**.

(1 x 2) (2)

(c) Give TWO impacts that the development of fog in this area could have on early morning traffic on the national route near **K**.

(2 x 1) (2)

3.2.2 Draw a cross-section along the line **F** - **G** across the Umgwenya River on the topographic map. Provide the following labels on the cross-section:

(i) Undercut slope
(ii) Slip-off slope

(1 + 2 x 1) (3)

3.2.3 Refer to point **J** on the topographic map. Explain why deposition is the main process taking place at this point.

(1 x 2) (2)

3.2.4 Refer to block **C2** on the topographical map.

(a) Identify the fluvial feature at **H** in block **C2** on the topographical map.

(1 x 1) (1)

(b) The feature identified in QUESTION 3.2.4 (a) is commonly found in the ... of a river where the river valley has a/an ...

(1 x 1) (1)

(i) upper course
(ii) middle course
(iii) U-shape.
(iv) V-shape.

A (i) and (iv)
B (i) and (iii)
C (ii) and (iii)
D (ii) and (iv)

3.3 GEOGRAPHIC INFORMATION SYSTEMS (GIS)

3.3.1 Refer to the dam in block **A5** on the topographical map.

(a) Is the dam an example of (vector/raster) data?

(1 x 1) (1)

(b) Give a reason for the answer to QUESTION 3.3.1 (a).

(1 x 2) (2)

3.3.2 (a) Define the term buffering.

(1 x 2) (2)

(b) Do you think buffering was applied in the development of the houses within Leopard Creek Golf Estate in block **C2**?

(1 x 1) (1)

(c) Give a reason from the topographic map for your answer to QUESTION 3.3.2 (b).

(1 x 2) (2)

[Question 3: 60 marks]

Section B Total: 30 marks

Exam Total: 150 marks